

Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

Code:

```
geya@LenovoL480: ~/OS/Pra × + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

#define SIZE 1000000

int main()
{
    int i;

    // Allocate memory before fork
    int *data = (int *)malloc(SIZE * sizeof(int));

    if (data == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    // Initialize the array
    for (i = 0; i < SIZE; i++)
    {
        data[i] = i;
    }
}
```

```

printf("Parent: Memory allocated and initialized.\n");
printf("Parent PID: %d\n", getpid());

pid_t pid = fork();

if (pid < 0)
{
    printf("fork failed\n");
    free(data);
    return 1;
}

if (pid == 0)
{
    // Child process
    printf("\nChild process created.\n");
    printf("Child PID: %d\n", getpid());
    printf("Child: data[0] before modification = %d\n", data[0]);

    // Modify data in child
    printf("Child: Modifying data...\n");

    for (i = 0; i < SIZE; i++)
    {
        data[i] = data[i] + 100;

```

```

        printf("Child: data[0] after modification = %d\n", data[0]);

        sleep(5);

        free(data);
        exit(0);
    }
    else
    {
        // Parent process
        printf("\nParent continues running.\n");
        printf("Parent: data[0] = %d\n", data[0]);

        sleep(2);

        printf("Parent: data[0] after child modification = %d\n", data[0]);

        wait(NULL);

        printf("\nChild process completed.\n");

        free(data);
    }
}

```

```
    return 0;  
}
```

Output:

```
geya@LenovoL480:~/OS/Practical/Week 8$ ./task2  
Parent: Memory allocated and initialized.  
Parent PID: 1354  
  
Parent continues running.  
Parent: data[0] = 0  
  
Child process created.  
Child PID: 1355  
Child: data[0] before modification = 0  
Child: Modifying data...  
Child: data[0] after modification = 100  
Parent: data[0] after child modification = 0  
  
Child process completed.  
geya@LenovoL480:~/OS/Practical/Week 8$ |
```